FARAJI S. MADUNDA

Statistician | Data Analyst | Survey & Poverty Analytics
Dar es Salaam, Tanzania | +255 620 219 947 | farajimadunda@gmail.com
linkedin.com/in/faraji-madunda-7928b11b3

PROFESSIONAL SUMMARY

Statistician and data analyst with a bachelor's degree in Official Statistics from the Eastern Africa Statistical Training Centre (EASTC). Works on national-level household survey analysis in Tanzania, with expertise in multidimensional poverty measurement (Alkire-Foster), survey sampling design, and geospatial modelling. Builds reproducible analysis workflows and interactive tools in Stata, R and Python and provides statistical advisory to development projects. Experienced in census field operations and training. Committed to evidence-based decision-making and global development.

CORE SKILLS

Statistical Methods: Household survey analysis, complex survey estimation and weighting, sampling design (stratified and cluster), sample size and power calculation, Multidimensional Poverty Index (Alkire-Foster), monitoring and evaluation (M&E), data quality assurance

Statistical Software & Programming: Stata, R (including INLA, openxlsx), Python (pandas, NumPy, SciPy, Plotly, GeoPandas), Streamlit, SPSS, CSPro

Geospatial Analysis: Spatial modelling with R-INLA, DHS GPS cluster data, administrative shapefiles, GeoPandas

Tools & Productivity: Microsoft Excel (pivot tables, lookups, logical functions), Word, PowerPoint, Outlook, Git and GitHub, VS Code, Jupyter, RStudio

Web: HTML, CSS, JavaScript (self-taught)

PROFESSIONAL EXPERIENCE

Statistician and Data Analyst **Feb 2025 - Present**

National Bureau of Statistics (NBS), Tanzania

- Analyse national household survey data, including the Tanzania Household Budget Survey (TIHBS), to produce national Multidimensional Poverty Index estimates by Mainland/Zanzibar, rural/urban and region, applying survey weights and survey-aware estimation in Stata.
- Construct deprivation indicators across housing, education, health, energy and financial access dimensions, and export results to Excel for reporting.
- Develop reproducible R and Python workflows for survey data management, including pipelines for the Comprehensive Enterprises Survey (CES) and the Tanzania National Panel Survey (TZNPS).
- Carried out geospatial analysis of immunization coverage in Simiyu Region using TDHS 2022 data, administrative shapefiles and point-referenced spatial models in R-INLA.

Statistical Advisor **Advisory engagement**

Advancing Gender Equality Project, Tanzania (World Bank-funded, 2023–2028)

- Advised on a two-stage stratified cluster sampling design targeting approximately 170 enumeration areas.
- Reviewed the M&E plan and identified risks of selection bias, absence of control groups and self-report bias; supported questionnaire variable development.
- Assessed TDHS 2022 as a substitute baseline data source.

Regional Content Supervisor **Jul 2022 – Sep 2022**

National Bureau of Statistics (NBS), Mtwara, Tanzania

- Oversaw district-level training for practice census studies and housing enumeration.
- Managed and directed trainers throughout the training period.
- Emphasised and ensured data accuracy during training and the population and housing counting exercise.

Data Collector **Feb 2022 – May 2022**

Ubungo Municipal Council, Dar es Salaam, Tanzania

- Delineated main and street roads in assigned areas and assigned road and street names.
- Entered household and compound information into the NaPA system.

- Performed additional tasks assigned by the supervisor.

Data Officer Assistant **Jul 2021 – Oct 2021**

Sinza Municipal Hospital, Dar es Salaam, Tanzania (Statistics Unit, Data Department)

- Entered routine records via DHS and analysed data in Excel.
- Prepared weekly and monthly reports and kept records organised.

SELECTED TECHNICAL PROJECTS

NMPI Automation Tool (Python, Streamlit)

- Multi-page application automating the Alkire-Foster workflow: data loading, dimension and indicator configuration, analysis, results viewing and export; redesigned with an improved user interface without changing any calculation logic.

Automated Sample Size & Survey Design Simulator (Python, Streamlit)

- Modules for simple random sampling, stratified and cluster designs (DEFF/ICC), power analysis and Monte Carlo simulation; validated against standard formulas, with CSV, Excel and PDF export.

National MPI Data Management System (R)

- Modular R project with an interactive workflow menu implementing the Alkire-Foster methodology and Excel output.

EDUCATION

Bachelor's Degree in Official Statistics **Aug 2018 – Jul 2021**

Eastern Africa Statistical Training Centre (EASTC), Dar es Salaam, Tanzania

Advanced Certificate of Secondary Education **Jan 2016 – May 2018**

Biharamulo Secondary School, Kagera, Tanzania

Certificate of Secondary Education **Jan 2012 – Oct 2015**

Mbezi Inn Secondary School, Dar es Salaam, Tanzania

Primary School Certificate **Jan 2006 – Oct 2011**

Makabe Primary School, Dar es Salaam, Tanzania

LANGUAGES

Swahili – Native | English – Professional working proficiency

INTERESTS

Reading, coding, building analytical tools